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Inequality 2

Directions (1 – 5): In the following questions the symbols #, *, @. $ and = are used with the following meanings:

A # B means A is greater than B.

A * B means A is greater than or equal to B.

A @ B means A is equal to B.

A $ B means A is lesser than B.

A = B means A is lesser than or equal to B.

Now in each of the following questions, assuming the three statements to be true, find which of the two conclusions I and II given below them is/are true. Give answer.

1) If only conclusion I is true

2) If only conclusion II is true

3) If either conclusion I or conclusion II is true

4) If neither conclusion I nor conclusion II is true

5) If both conclusions I and II are true.

1. Statements: Q # R, S $ Q, S * P

Conclusions: I. R # S   II. R $ S

Ans: 4

Decoded statement: Q > R, S < Q, S ≥ P

Decoded conclusion: I. R ≤ S   II. R < S

Combined Inequalities: P ≤ S < Q > R

P ≤ S < Q > R   No relationship can be established between R and S. Hence conclusion I and II does not follow.

2. Statements: A = B, D @ C, C # A

Conclusions: I. D = B   II. C * D

Ans: 4

Decoded statement: A ≤ B, D = C, C > A

Decoded conclusion: I. D ≤ B II. C ≥ D

Combined Inequalities: D = C > A ≤ B

D = C > A ≤ B No relationship can be established between D and B. Hence conclusion I does not follow.

D = C > A ≤ B D = C. Hence conclusion II does not follow.

3. Statements : Q @ R, M @ N, Q # M

Conclusions : I. R # N II. N $ Q

Ans: 5

Decoded statement: Q = R, M = N, Q > M

Decoded conclusion: I. R > N II. N < Q

Combined Inequalities: N = M < Q = R

N = M < Q = R R > N. Hence conclusion I follows.

N = M < Q = R N < Q. Hence conclusion II follows.

4. Statements: Q # N # M, M # O @ R, R $ T @ S

Conclusions : I. S @ N II. M @ S

Ans: 4

Decoded statement: Q > N > M, M > O = R, R < T = S

Decoded conclusion: I. S = N II. M = S

Combined Inequalities: Q > N > M > O = R < T = S

Q > N > M > O = R < T = S No relationship can be established between S and N. Hence conclusion I does not follow.

Q > N > M > O = R < T = S No relationship can be established between M and S. Hence conclusion II does not follow.

5. Statements : Q * R, R @ U, U * M

Conclusions : I. R # M   II. M @ R

Ans: 3

Decoded statement: Q ≥ R, R = U, U ≥ M

Decoded conclusion: I.   R > M   II. M = R

Combined Inequalities: Q ≥ R = U ≥ M

Q ≥ R = U ≥ M   R ≥ M. Hence either conclusion I or II follows.

Directions (Q. 6-10) In the following questions, the symbols *, +, $, #, δ are used with the following meanings.

A * B means 'A is not smaller than B'.

A + B means 'A is neither greater than nor smaller than B'.

A $ B means 'A is not greater than B'.

A # B means 'A is neither smaller than nor equal to B'.

A δ B means 'A is neither greater than nor equal to B'.

Now in each of the following questions, assuming the given statements to be true, find which of the two conclusions I and II given below them is/are definitely true.

Give answer

1) if only conclusion I is true.

2) if only conclusion II is true.

3) if either conclusion I or II is true.

4) if neither conclusion I nor II is true.

5) if both conclusions I and II are true.

6). Statements: L * I, F # G, L + G

Conclusions: I.F # I   II. I δ G

Ans: 1

Decoded statement: L ≥ I, F > G, L = G

Decoded conclusion: I. F > I II. I ≥ G

Combined Inequalities: F > G = L ≥ I

F > G = L ≥ I F > I. Hence conclusion I is true.

F > G = L ≥ I G ≥ I. Hence conclusion II is not true.

7). Statements: H # R, Z δ Q, Q $ Y, R # Y

Conclusions: I.R # Q II.H # Z

Ans: 5

Decoded statement: H > R, Z < Q, Q ≤ Y, R > Y

Decoded conclusion: I. R > Q II. H > Z

Combined Inequalities: H > R > Y ≥ Q > Z

H > R > Y ≥ Q > Z R > Q. Hence conclusion I is true.

H > R > Y ≥ Q > Z H > Z. Hence conclusion II is true.

8). Statements: E + Q, N * E, Q * X

Conclusions: I.N + X II.N # X

Ans: 3

Decoded statement: E = Q, N ≥ E, Q ≥ X

Decoded conclusion: I. N = X II. N > X

Combined Inequalities: N ≥ E = Q ≥ X

N ≥ E = Q ≥ X N ≥ E. Hence either conclusion I or II is true.

9). Statements: K * V, B $ W, W * K

Conclusions: I.B + K II.W + V

Ans: 4

Decoded statement: K ≥ V, B ≤ W, W ≥ K

Decoded conclusion: I. B = K II. W = V

Combined Inequalities: B ≤ W ≥ K ≥ V

B ≤ W ≥ K ≥ V No relationship cam be established between B and K. Hence conclusion I is not true.

B ≤ W ≥ K ≥ V No relationship cam be established between W and V. Hence conclusion II is not true.

10). Statements: P # Q, R + P, Q $ M

Conclusions: I.R # M II.Q + M

Ans: 4

Decoded statement: P > Q, R = P, Q ≤ M

Decoded conclusion: I. R > M II. Q = M

Combined Inequalities: R = P > Q ≤ M

R = P > Q ≤ M No relationship can be established between R and M. Hence conclusion I is not true.

R = P > Q ≤ M Q ≤ M. Hence conclusion II is not true.

Directions (11-15): In the following questions, the symbols %, &, #, * and @ are used with the following meaning as illustrated below-

‘P#Q’ means ‘P is neither greater than nor equal to Q’

‘P*Q’ means ‘P is neither equal to nor smaller than Q’

‘P%Q’ means ‘P is neither smaller than nor greater than Q’

‘P@Q’ means ‘P is not smaller than Q’

‘P&Q’ means ‘P is not greater than Q’

Now in each of the following questions assuming the given statement to be true, find which of the conclusions given below them is/are definitely true and give your answer accordingly.

Q11. Statements: D * U @ W # S; U * Q % Z @ P

Conclusions: I. D * P II. S @ Z

None is true

Only I is true

Only II is true

Either I or II is true

Both are true

Ans: 2

Decoded statement: D > U ≥ W < S, U > Q = Z ≥ P

Decoded conclusion: I. D > P II. S ≥ Z

D > U > Q = Z ≥ U D > P. Hence conclusion I is true.

Z = Q < U ≥ W < S No relationship can be established between S and Z. Hence conclusion II is not true.

Q12. Statements: J & L & O % R * X @ D * Q % V

Conclusions: I. R * J II. R % J

None is true

Both are true

Only II is true

Either I or II is true

Only I is true

Ans: 4

Decoded statement: J ≤ L ≤ O = R > X ≥ D > Q = V

Decoded conclusion: I. R > J II. R = J

J ≤ L ≤ O = R > X ≥ D > Q = V J ≤ R. Hence either conclusion I or II is true.

Q13. Statements: P @ B # S % T & N; M * N % D & A

Conclusions: I. A * B II. T * P

None is true

Only I is true

Only II is true

Either I or II is true

Both are true

Ans: 2

Decoded statement: P ≥ B < S = T ≤ N, M > N = D ≤ A

Decoded conclusion: I. A > B II. T > P

P ≥ B < S = T ≤ N = D ≤ A A > B. Hence conclusion I is true.

P ≥ B < S = T ≤ N No relationship can be established between T and P. Hence conclusion II is not true.

Q14. Statements: X & W % D % M * U * A # I

Conclusions: I. X # U II. I * W

None is true

Only I is true

Only II is true

Either I or II is true

Both are true

Ans: 1

Decoded statement: X ≤ W = D = M > U > A < I

Decoded conclusion: I. X < U II. I > W

X ≤ W = D = M > U > A < I No relationship can be established between X and U. Hence conclusion I is not true.

X ≤ W = D = M > U > A < I No relationship can be established between I and W. Hence conclusion II is not true.

Q15. Statements: E @ N & A * L # Q; E * M * O % Y

Conclusions: I. Y # N II. A * Y

None is true

Only I is true

Only II is true

Either I or II is true

Both are true

Ans: 1

Decoded statement: E ≥ N ≤ A > L < Q, E > M > O = Y

Decoded conclusion: I. Y < N II. A > Y

Y = O < M < E ≥ N ≤ A > L < Q No relationship can be established between Y and N. Hence conclusion I is not true.

Y = O < M < E ≥ N ≤ A > L < Q No relationship can be established between A and Y. Hence conclusion II is not true.

Directions (16-20): Study the following information carefully to answer the given questions.

P @ Q – P is neither greater than nor equal to Q

P % Q – P is neither smaller than nor equal to Q

P # Q – P is not greater than Q

P $ Q – P is not smaller than Q

P * Q- P is neither smaller than nor greater than Q

16. Statements: A @ Z, Z % Y, Y * X, X $ W

Conclusions: I. Z % W II. A % W

Only conclusion I follows

Only conclusion II follows

Either conclusion I or II follows

Neither conclusion I nor II follow

Both conclusion I and II follows

Ans: 1

Decoded statement: A < Z, Z > Y, Y = X, X ≥ W

Decoded conclusion: I. Z > W II. A > W

A < Z > Y = X ≥ W Z > W. Hence conclusion I is true.

A < Z > Y = X ≥ W No relationship can be established between A and W. Hence conclusion II is not true.

17. Statements: P % S, Q @ T, S # R, T * R

Conclusions: I. R $ P II. T @ P

Only conclusion I follows

Only conclusion II follows

Either conclusion I or II follows

Neither conclusion I nor II follow

Both conclusion I and II follows

Ans: 4

Decoded statement: P > S, Q < T, S ≤ R, T = R

Decoded conclusion: I. R ≥ P II. T < P

P > S ≤ R = T > Q No relationship can be established between R and P. Hence conclusion I is not true.

P > S ≤ R = T > Q No relationship can be established between P and T. Hence conclusion II is not true.

18. Statements: H * L, L $ Y, Y # Q, Q @ O

Conclusions: I. L % Q II. Y @ O

Only conclusion I follows

Only conclusion II follows

Either conclusion I or II follows

Neither conclusion I nor II follow

Both conclusion I and II follows

Ans: 2

Decoded statement: H = L, L ≥ Y, Y ≤ Q, Q < O

Decoded conclusion: I. L > Q II. Y < O

H = L ≥ Y ≤ Q < O No relationship can be established between L and Q. Hence conclusion I is not true.

H = L ≥ Y ≤ Q < O Y < O. Hence conclusion II is true.

19. Statements: D $ E, H $ F, E * H, F @ G

Conclusions: I. D $ F II. G @ E

Only conclusion I follows

Only conclusion II follows

Either conclusion I or II follows

Neither conclusion I nor II follow

Both conclusion I and II follows

Ans: 1

Decoded statement: D ≥ E, H ≥ F, E = H, F < G

Decoded conclusion: I. D ≥ F II. G < E

D ≥ E = H ≥ F < G D ≥ F. Hence conclusion I is true.

D ≥ E = H ≥ F < G No relationship can be established between G and E. Hence conclusion II is not true.

20. Statements: J % K, M $ N, L @ K, L * M

Conclusions: I. J % M II. N @ K

Only conclusion I follows

Only conclusion II follows

Either conclusion I or II follows

Neither conclusion I nor II follow

Both conclusion I and II follows

Ans: 5

Decoded statement: J > K, M ≥ N, L < K, L = M

Decoded conclusion: I. J > M II. N < K

J > K > L = M ≥ N J > M. Hence conclusion I is true.

J > K > L = M ≥ N K > N. Hence conclusion II is true.